

Bangladesh University of Engineering and Technology (BUET)
Department of Industrial and Production Engineering (IPE)



Course No: IPE 6107
Course Title: Machine learning I

Holistic Framework for Bangla Sentiment Analysis: Accuracy, Efficiency, and Sustainability

Submitted to
Dr. Ridwan Al Aziz
Assistant Professor, Department of IPE, BUET

Submitted by

Team No – 15

Mostafizur Rahman Khan (Team Leader)	ID- 0424082030
Nayan Nath	ID- 0424082027
Sourav Paul	ID- 0423082006
Sanjida Hassan	ID- 0423082116

Table of Contents

Abstract	2
1 Introduction.....	3
2 Literature Review.....	6
2.1 Current Knowledge	6
2.2 Literature Review Table.....	7
2.3. Research Gap.....	16
2.4. Research Objectives and Questions	17
3. Methodology	18
3.1 Data Extraction and Preprocessing	19
3.2 Topic Modeling	20
3.3 Sentiment Classification.....	23
3.4 Comparative Framework for Sentiment Analysis	27
3.4.1 Classical Topic Modeling + Encoders.....	27
3.4.2 LLM-based Topic Modeling + Encoders	27
3.4.3 Direct LLM Prompting	28
3.5 Evaluation Metrics	29
3.5 Robust Optimization Formulation.....	30
3.6 Trade-off Analysis.....	33
4. Data collection plan	33
5. Conclusion	33
5.1 Summary	33
5.2 Limitations	34
6. Reference	35

Abstract

The rapid spread of misinformation and rumors on social media poses significant societal and economic risks, particularly in low-resource language contexts such as Bangla. While sentiment analysis and topic modeling have been extensively studied in English and other high-resource languages, research on Bangla remains limited due to linguistic complexities, scarce datasets, and inadequate tools. This study proposes a holistic framework for Bangla sentiment prediction that integrates topic modeling with sentiment classification while incorporating efficiency and sustainability into evaluation. Specifically, three comparative paradigms are investigated: (i) classical embedding-based topic modeling with encoders (BanglaBERT, XLM-RoBERTa), (ii) large language model (LLM)-based topic modeling combined with encoders, and (iii) direct LLM prompting. A benchmark of 36 model configurations is established, enabling systematic evaluation across accuracy, interpretability, modularity, scalability, and contextual adaptability. Unlike prior studies that emphasize accuracy alone, this work introduces a robust optimization-based evaluation framework that accounts for latency, energy consumption, carbon footprint, and cost-sensitive misclassification errors, ensuring robustness under uncertain deployment conditions. Data for the study is drawn from Bangla rumor-related Facebook posts and comments, collected via a semi-automatic pipeline and preprocessed through normalization, tokenization, and code-mixing resolution. The framework highlights trade-offs between traditional lightweight models and computationally intensive LLMs, revealing that BanglaBERT performs well for Bangla-specific tasks, while LLM-based approaches provide richer embeddings but incur higher energy and latency costs. Ultimately, this study contributes both methodological innovation and practical guidelines by bridging accuracy-driven benchmarks with cost-aware and sustainable evaluation, offering a comprehensive foundation for sentiment analysis in low-resource languages.

Keywords: Social media, Sentiment Analysis, Topic Modeling, LLMs, Robust Optimization.

1 Introduction

Social media platforms now dominate online activity, enabling rapid viral spread of even minor information and fostering relationships among users. This makes them powerful yet vulnerable channels for misinformation and rumors. In 2021, Facebook reported removing over 12 million posts that contained misinformation about COVID-19 and vaccines (Lin & Chen, 2021). In recent times, following the shooting of conservative activist Charlie Kirk in Utah, USA, false claims and conspiracy theories about the shooter's political affiliations and foreign involvement spread widely on Facebook, further fueling misinformation and political polarization (Politico, 2025). In Bangladesh, over false child-abduction claims linked to a mega-bridge project, unchecked social media rumors influenced public opinion, disrupted markets, and fueled vigilante violence that left at least eight people dead (Paul & Yesmin, 2024). Thus, the rapid spread of fake news on social networks has become a pressing worldwide issue, and to prevent social panic and potential economic disruption, swift action is required.

A crucial part of addressing rumors involves examining the main themes in rumor-driven discussions and understanding the emotional responses they provoke. Identifying the topics that arise in such conversations can reveal patterns in how rumors spread, and sentiment analysis can provide insights into public perception and emotional reactions. This involves automatically evaluating texts to determine the expressed positive or negative opinions. The most common task, sentiment classification, categorizes subjective texts emotionally, typically as positive, negative, or sometimes neutral (Wang et al., 2021). Sentiment analysis, also known as opinion mining, has gained increasing attention in recent years. Early overviews, such as Ligthart et al. (2021), examined opinion mining at its initial stages, while Piryani et al. (2017) reviewed studies from 2000 to 2015 and proposed frameworks for processing unstructured data to extract opinions and emotions. Recent surveys (Yousif et al., 2019; Birjali et al., 2021) discuss key challenges and potential directions in sentiment analysis, including sentiment classification (Yadav & Vishwakarma, 2020) and the detection of opinion spam and fraudulent reviews (Jain et al., 2021). Additionally, studies have explored the effectiveness of online reviews and the application of machine learning in social media analysis (Balaji et al., 2021). Sentiment analysis commonly employs a variety of approaches, including pre-trained models, traditional machine learning (ML)

techniques, and custom deep learning (DL) architectures. Pre-trained models such as VADER, TextBlob, and Flair are widely used for classifying text as positive, negative, or neutral, benefiting from extensive language knowledge bases and specialized preprocessing (Hutto & Gilbert, 2014; Loria, 2018; Akbik et al., 2019). Traditional ML models, including logistic regression, support vector machines, k-nearest neighbors, decision trees, random forest, and Naive Bayes, are often applied to explore classification performance across different datasets (Mitra & Mohanty, 2020). Advanced Transformer-based models like BERT and GPT-2 leverage contextual understanding and large-scale pre-training (Devlin et al., 2019; Mathew & Bindu, 2020), while custom DL models, such as CNNs, RNNs, and Bidirectional LSTM networks, are used to capture sequential patterns and long-term dependencies in text (Gandhi et al., 2021; Pennington et al., 2014).

Although sentiment analysis has seen significant progress for widely spoken languages like English, its application to low-resource languages such as Bengali remains limited. Recent developments, including transformer-based models like BanglaBERT and XLM-RoBERTa, have enhanced sentiment classification in Bengali (Islam et al., 2020; Hoque & Siddiqui, 2024), yet large-scale, publicly available datasets and tools for nuanced sentiment analysis in Bengali social media remain scarce (Hoque & Siddiqui, 2023). The linguistic complexities of Bengali, including informal expressions, code-mixing, sarcasm, and cultural context, further challenge automated analysis. It is widely recognized that social media platforms serve not only as channels for sharing information but also as spaces for exchanging opinions, expressing ideologies, and engaging emotionally. In Bangladesh, rumors on platforms like Facebook have had real-world consequences, emphasizing the importance of analyzing both the topics and emotional responses associated with these discussions. Considering Bangladesh's high social media engagement (Amin et al., 2024), developing Bengali-specific sentiment analysis tools is crucial for detecting and mitigating misinformation. However, most existing research focuses on English and other high-resource languages, leaving a gap in understanding rumor dynamics in Bengali social media. This study seeks to address this gap by combining topic modeling and sentiment analysis to examine the themes and emotional trends in Bengali rumor discussions.

Several studies have conducted multi-class sentiment analysis in Bengali using a range of machine learning (ML) and deep learning (DL) models. Haque et al. (2023) applied CNN and LSTM-based classifiers to Bengali Facebook comments, achieving up to 85.8% accuracy and 86% F1 scores.

Das et al. (2019) combined encoder-decoder LSTM networks with TF-IDF, Word2Vec, and 1D CNN for hateful language detection. Various ML algorithms, including Random Forest, multinomial Naive Bayes, and SVM with multiple kernels, were evaluated for classification tasks (Eshan & Hasan, 2017), while GRU networks categorized social media feedback into multiple sentiment classes (Ishmam & Sharmin, 2019). Neural network ensembles were employed for binary and multi-label abuse detection (Ahmed et al., 2021), and ML/DL methods were applied to detect cyberbullying in Bangla and Romanized Bangla texts (Ahmed et al., 2021). Transformer-based models such as Bangla BERT, DistilBERT, and XLM-RoBERTa were tested for Bengali comment classification, with XLM-RoBERTa achieving the highest accuracy (Emon et al., 2019).

This study investigates the spread of rumors in Bengali social media by integrating two key approaches: topic modeling and sentiment analysis. The sentiment prediction in Bangla text is explored through a two-stage strategy that balances explainability with modern scalability. The first stage combines topic modeling with encoder-based sentiment analysis, while the second relies on large language models (LLMs) for prompting-based sentiment classification. Within these strategies, different configurations of topic models, encoders, and LLMs are systematically compared. Table 8 in the *Methodology* section exhibits the combinations for sentiment analysis. In this study, we move beyond the conventional evaluation of sentiment analysis models that relies solely on statistical metrics such as accuracy, precision, recall, F1-score, and confusion matrices. While these measures are essential for assessing predictive quality, they overlook the practical and operational costs of real-world deployment. For example, some studies estimate that training GPT-3 on a 500-billion-word dataset consumed approximately 1,287 MWh of electricity and generated about 550 tons of CO₂ emissions (Patterson et al., 2021). Moreover, Sagi (2025) emphasized the importance of evaluating and balancing key inference metrics—such as latency—to optimize the responsiveness and efficiency of large language models in real-time applications. To address these limitations, we introduce a robust, optimization-based framework that integrates multiple dimensions of cost into the evaluation process. Specifically, the framework considers the latency of prediction, the energy consumed during inference, and the resulting carbon emissions, alongside a cost-sensitive error term that accounts for the varying severity of different misclassifications. For example, in sentiment classification, incorrectly predicting a negative opinion as positive may carry a much higher cost than confusing a positive sentiment with neutral. Furthermore, by adopting a robust min-max formulation, we explicitly model the uncertainty in these coefficients,

ensuring that the chosen model remains effective under worst-case scenarios, such as fluctuating electricity prices, dynamic carbon taxation, or shifting business priorities. This holistic evaluation allows us to assess models not only in terms of predictive performance but also in terms of time efficiency, sustainability, and error impact, thereby bridging the gap between machine learning metrics and operational decision-making.

The structure of this paper is organized as follows. Section 2 reviews existing literature on sentiment analysis, including studies on English texts, and multilingual sentiment analysis, while identifying key research gaps. Section 3 describes the proposed analytical framework, detailing the topic modeling and sentiment analysis models, along with the evaluation criteria used in this study. Section 4 presents the data collection plan. Finally, Section 5 concludes the study by summarizing the key insights on sentiment analysis in low-resource languages and outlining the anticipated limitations.

2 Literature Review

This section critically examines existing research on sentiment analysis, emphasizing its application in low-resource languages, the role of topic modeling, and the growing relevance of sustainability in evaluating models. By synthesizing insights from recent studies, it builds a foundation for the present work while highlighting the key gaps this research seeks to address.

2.1 Current Knowledge

Sentiment analysis has evolved from lexicon-based and traditional machine learning methods (e.g., SVM, Naive Bayes) to sophisticated deep learning and Large Language Model (LLM)-based approaches (Van & Bleich, 2025; Pandya, 2024). The emergence of transformer-based models, notably BERT (Yates et al., 2021) and multilingual versions like XLM-RoBERTa (Gaikwad et al., 2023), has greatly enhanced the capacity to capture contextual meaning in text. For linguistically specific tasks, monolingual models like BanglaBERT have demonstrated superior performance by capturing language-specific syntax and semantics more effectively than their multilingual counterparts (Kowsher et al., 2022).

Concurrently, topic modeling techniques have advanced from probabilistic models like Latent Dirichlet Allocation (LDA) to embedding-based methods such as BERTopic (Grootendorst, 2022). These modern techniques leverage contextual embeddings to generate more coherent and semantically meaningful topics, which can be highly beneficial for preprocessing and enriching text data for downstream tasks like sentiment analysis.

The integration of LLMs has further expanded the frontier. Studies have shown that LLMs like GPT-4, Llama, and Qwen can perform zero-shot or few-shot sentiment analysis and topic modeling with remarkable accuracy, even on low-resource languages through cross-lingual transfer (Miah et al., 2024; Wahidur et al., 2024). However, this performance often comes at a significant computational cost. The predominant focus in the literature remains on maximizing accuracy metrics (e.g., F1-score, accuracy), while practical deployment constraints such as inference latency, energy consumption, and carbon emissions are largely overlooked (Różycki et al., 2025).

2.2 Literature Review Table

A systematic review of recent literature (2023-2025) was conducted to map the current landscape. The table below summarizes key studies, Data collection platform, Data collection way, Study population, Considerable Features, Applied Tools, Area of application, Objective of the study, Results, and their limitations.

Table 1: Data collection information, features, and tools used in recent studies

No.	Article	Data collection platform	Data collection way	Study population	Considerable Features	Applied Tools
1	Xing (2025)	News and social media platforms	Six FSA datasets (FPB, StockSen, CMC, FiQA, SEntFiN and FinEntity)	32835 datapoints	Financial news, social media posts and forum discussions	Heterogeneous LLM agents framework, summative agent reasoning
2	Katta (2025)	Reddit	AsyncPRAW (Asynchronous Python	10733 entries	Reddit comments	Sentiment analysis, LDA (Latent

			Reddit API Wrapper) library			Dirichlet Allocation), word frequency approach, topic modeling
3	Invernici et al. (2025)	Academic literature	Scopus database	Hundreds of thousands of documents (scientific abstracts, 2006-2023)	Abstracts and their metadata	BERTopic, LLM-based embeddings, hyperparameter optimizer, TETYS dashboard
4	Wahidur et al. (2024)	Social media platforms	Neo dataset, Bitcoin sentiment dataset, Reddit dataset and Cryptocurrency sentiment dataset	14091 tweets	Sentiment, tweet, posts	Microsoft's MiniLM (distilled BERT), Google's FLAN-T5, instruction-based fine-tuning
5	Uddin & Sarker (2024)	Real-world phishing email dataset	Sourced from Kaggle website.	11,322 legitimate Emails and 7,328 phishing Emails	Email content, class imbalance, positive and negative contribution scores	RoBERTa (fine-tuned), LITA (LIME-Transformer Attribution), LIME, Transformers Interpret
6	Xu et al.(2024)	MIMIC-CXR dataset	Not mentioned	500 radiology reports	Semantic similarity, text labels, clinical ground truth alignment	GPT-4, zero-shot learning framework
7	Ghatora et al. (2024)	Customer product reviews	Sourced from Kaggle website.	205,052 records	Product name, product price, rate, review, summary and sentiment	Random Forest, Naive Bayes, SVM, GPT-4

					(positive, negative and neutral).	
8	Thakur et al. (2024)	YouTube (with cross-platform analysis on Facebook and Twitter)	YouTube video descriptions analysis	8122 videos	Four distinct themes: Promotion/Outreach, Treatment, Conspiracy Theories, Politics; sentiment distribution, language diversity, sharing patterns	Topic modeling, topic-specific sentiment analysis, language detection, correlation analysis
9	Miah et al. (2024)	Not specified	SemEval-2017 Task 4: Sentiment Analysis in Twitter, amazon_reviews_multi, DEFT 2017, and SENTIPOLC 2016	8375 datapoints	Sentiment polarity (positive, negative, neutral), cross-lingual analysis	LibreTranslate, Google Translate, Twitter-Roberta-Base-Sentiment-Latest, bert-base-multilingual-uncased-sentiment, GPT-3
10	Rehan et al. (2023)	Twitter and YouTube	Not mentioned	Urdu 2400 tweets and english 1313 posts	Tweets and comments	MuRIL, RoBERTa, transfer learning, fine-tuning, manual search and grid search for hyper-parameters
11	Khan et al. (2025)	Not mentioned	Case study and experimental framework with quantization techniques	Multiple LLMs: Llama3.2, Phi3.2, Mistral,	Energy consumption, carbon emissions, model accuracy, precision,	4-bit quantization, Ollama platform, local inference optimization,

			and local inference optimization	Qwen, Llava with parameter counts from 1B to 32B	recall, F1-score, quantization effects	energy consumption monitoring
12	Sagi (2025)	Not mentioned	Literature review and performance analysis of LLM inference metrics across different deployment scenarios	Various LLM configurations including GPT-2 (124M) to LLaMA-2 (13B) parameters across different hardware setups	Time to First Token (TTFT), generation time, end-to-end latency, inter-token latency, tokens per second (TPS), requests per second (RPS)	Performance benchmarking tools, latency measurement frameworks, throughput analysis, system profiling
13	Wang et al. (2025)	Not mentioned	Experimental framework with concurrency optimization using speculative decoding and parallel branches across multiple model sizes	Models from 3B to 32B parameters including s1.1-32B, LLaMA-3.1-8B, QwQ-32B, tested on MATH-500, AIME24/25, GPQA-Diamond	Token efficiency, throughput optimization, branch-wise parallelism, sequence-wise parallelism, latency constraints	Speculative decoding, majority voting, greedy search algorithm, Ollama deployment, Eagle3 draft models
14	Iftikhar & Davy (2024)	Not mentioned	Framework development with federated learning approach and renewable energy source prioritization	Global manufacturing company use case with multiple facilities using federated	Carbon emissions, energy consumption, federated learning efficiency, renewable energy integration, data privacy preservation	Federated learning coordinator (FLC), Parameter Efficient Fine Tuning (PEFT), renewable energy source

				learning network		selection algorithm
15	Liu & Yin (2024)	Not mentioned	Experimental study using CodeCarbon's CO2 emissions tracker during model training	Large Language Models: BERT, DistilBERT, and T5 with different tokenizers	Power consumption (kWh), CO2 emissions, cloud compute usage, training time, hardware efficiency, renewable energy usage, hyperparameter sensitivity	CodeCarbon CO2 emissions tracker, Power monitoring tools, GPU performance analysis (T4 vs A100), Model training frameworks, Loss calculation, Cosine similarity measurement, STS scoring

Table 2: Areas, objectives, results, and limitations of recent studies

No.	Article	Area of application	Objective of the study	Results	Limitations
1	Xing (2025)	Miscellaneous	Investigate the effectiveness of using LLMs without fine-tuning for financial sentiment analysis through heterogeneous LLM agents framework	Better accuracies compared to alternative multi-LLM agent settings, especially when discussion contents are substantial.	Since LLM-based prediction is slower and costlier than statistical analysis, this makes large multi-agent systems less practical.
2	Katta (2025)	Socialization	Analyze Reddit discussions about	Identified significant topics including trust in	The analysis is restricted to

			ChatGPT and DeepSeek using sentiment and topic modeling to understand user attitudes and public perception of LLMs	AI, user expectations, potential uses, reservations about AI biases, and ethical implications. Provides insights into public sentiment shaping AI development direction.	Reddit discussions.
3	Invernici et al. (2025)	Teaching and Learning	Propose a completely automated pipeline to uncover discussions on SDGs within research literature using topic modeling techniques	Successfully identified hundreds of topics on hundreds of thousands of documents. Captured insights on evolution of attitude toward SDGs in scientific abstracts (2006-2023). Produced interactive visualizations of topics' temporal evolution.	Topic modeling in a high-dimensional latent space may yield overlapping topics.
4	Wahidur et al. (2024)	Miscellaneous	To improve sentiment analysis for cryptocurrency market prediction by testing fine-tuning methods on DistilBERT, MiniLM, and FLAN-T5, focusing on instruction type, model size, and data scale.	40% average zero-shot performance gain after fine-tuning. Larger models achieved 75.16% accuracy. Short instructions: 72.38% accuracy (12% better than complex instructions).	Smaller-scale models may experience reduced generalization due to complete model capacity utilization.
5	Uddin & Sarker (2024)	Miscellaneous	Present a fine-tuned transformer-based language model (RoBERTa) for detecting phishing emails and propose a hybrid explanation approach for model transparency	Fine-tuned model achieves high accuracy in phishing email classification. LITA provides user-friendly insights and solves disagreement issues between LIME and	Lack of multilingual and phishing email datasets.

				Transformers Interpret approaches.	
6	Xu et al.(2024)	Health Care	Leverage LLM to enhance semantic analysis and develop similarity metrics for texts, addressing limitations of traditional unsupervised NLP metrics	GPT-4 generated labels significantly improve semantic similarity assessment with scores more closely aligned with clinical ground truth than traditional NLP metrics (ROUGE, BLEU).	Framework tested only on radiology reports; generalizability to other specialized domains needs validation.
7	Ghatora et al. (2024)	Miscellaneous	Explore the application of traditional machine learning and pre-trained LLMs for automated sentiment analysis of customer product reviews	Traditional models (especially SVM) better for short texts. GPT-4 showed superior performance with detailed texts, capturing subtle sentiments with higher precision, recall, and F1 scores. GPT-4 uniquely identified mixed sentiments not found in simpler models.	This research relied on a public dataset, which may not reflect the nuances of private organizational data.
8	Thakur et al. (2024)	Socialization	Analyze misinformation about COVID-19 on YouTube through topic modeling, sentiment analysis, language analysis, and cross-platform sharing patterns	4 themes identified. Sentiment analysis: Promotion (45.8% neutral, 39.8% positive); Treatment (38.1% positive, 31.3% neutral); Conspiracy (46.9% positive, 31.0% neutral); Politics (35.7% positive, 32.9% negative). English videos shared most on Facebook/Twitter. Significant correlation between video title	May not capture the full scope of misinformation dynamics across other social media platforms such as Facebook or TikTok.

				length and social media shares.	
9	Miah et al. (2024)	Miscellaneous	Propose an ensemble model of transformers and LLM for sentiment analysis of foreign languages by translating them into English	Over 86% accuracy in sentiment analysis on translated sentences. Ensemble model outperformed independent pre-trained models and LLM individually	Challenging when dealing with foreign languages without labelled training data.
10	Rehan et al. (2023)	Socialization	Address Multi-lingual Threatening Content Detection (MTCD) using transfer learning with fine-tuning techniques	RoBERTa achieved highest performance: 92% accuracy and 90% macro F1-score with joint multi-lingual approach.	Very limited work in low-resource languages.
11	Khan et al. (2025)	LLM Optimization & Energy Efficiency	Explore integration of energy-efficient optimization techniques in LLM deployment through quantization and local inference to reduce carbon footprint while maintaining performance	Up to 45% reduction in energy consumption and carbon emissions post-quantization. Quantized models maintain comparable accuracy while significantly reducing computational overhead.	Performance trade-offs in some metrics. Hardware dependency for local inference efficiency. Limited evaluation on diverse datasets needed.
12	Sagi (2025)	LLM Inference Performance Metrics	Provide comprehensive analysis of performance metrics for LLM inference in real-time applications, focusing on latency optimization and system efficiency	TTFT increases with model size (90-120ms for GPT-2 to 600-1000ms for LLaMA-2). TPS varies from 20-100 depending on model complexity. Hardware optimization shows significant impact on inference efficiency.	Limited to specific model architectures. Hardware-specific optimizations not universally applicable. Need for standardized

					benchmarking protocols.
13	Wang et al. (2025)	Latency-Aware Test-Time Scaling	Introduce latency-aware test-time scaling for LLMs, optimizing concurrency configurations through branch-wise and sequence-wise parallelism for latency-optimal performance	32B model achieves 82.3% accuracy on MATH-500 in 1 minute. 3B model reaches 72.4% in 10 seconds. Optimal concurrency provides up to 7.3% performance improvement with 1.7× latency reduction.	Memory-bound assumption limited to small/medium scale hardware. Limited to small batch scenarios. Performance trade-offs in compute-bound regimes.
14	Iftikhar & Davy (2024)	Sustainable AI Training Framework	Present energy-efficient training framework of LLMs employing federated learning and renewable energy integration to mitigate environmental impact	Framework enables carbon footprint reduction through RE prioritization. PEFT reduces training parameters by 95% (from 175B to ~7.9B parameters). Demonstrates 70 tonnes CO2 reduction potential for OPT model training.	Limited to federated learning scenarios. Requires infrastructure with renewable energy availability. Framework validation needed on larger scale deployments.
15	Liu & Yin (2024)	AI Energy Consumption Policy	To evaluate CO2 emissions of well-known large language models, propose strategies to reduce carbon emissions, and examine the impact of different hardware choices on CO2 emissions during model training	A100 GPU reduced training time by 62.6% and CO2 emissions by 83% compared to T4 GPU. DistilBERT with distilbert-base-uncased tokenizer had 46.9% fewer CO2 emissions than BERT with bert-base-cased tokenizer. T5	Study focuses only on CO2 emissions, not comprehensive environmental impact. Limited to specific models (BERT, DistilBERT, T5). Hardware

				model showed best validation loss but highest emissions on T4, significant improvement on A100. Models can maintain performance while reducing emissions through hardware optimization and model selection	costs may limit accessibility (A100 GPU costs \$6,799 vs T4 at \$1,176). Geographic and temporal variations in energy sources not fully accounted for. Limited evaluation framework - doesn't include sequence labeling tasks
--	--	--	--	--	---

2.3. Research Gap

The synthesis of the current literature reveals several critical gaps that this research seeks to bridge:

1. **Lack of a Holistic Bangla-Specific Framework:** There is a absence of systematic studies that combine traditional and LLM-based topic modeling with encoder models specifically for Bangla sentiment analysis. The relative performance trade-offs between these hybrid pipelines and direct LLM prompting remain unexplored.
2. **Neglect of Sustainability and Efficiency:** Prior research predominantly prioritizes accuracy, failing to incorporate crucial real-world deployment metrics such as inference latency, energy consumption, and carbon footprint. This omission makes it difficult to assess the true practicality and environmental impact of proposed models.
3. **Insufficient Exploration for Low-Resource Languages:** While multilingual LLMs (e.g., Llama, Qwen) and translation models (e.g., M2M100) offer potential for processing Bangla text, their efficacy and cost-effectiveness compared to dedicated monolingual encoders like BanglaBERT are not adequately benchmarked.

4. **Absence of Robust Optimization:** Existing evaluation frameworks do not account for operational uncertainties, such as fluctuating computational costs and energy prices. A robust optimization approach is needed to identify models that perform reliably under varying constraints.

2.4. Research Objectives and Questions

To address the identified gaps, this study proposes the following research questions (RQs):

RQ1: How can traditional topic modeling (LDA, BERT-LDA, BERTopic) and LLM-based topic modeling be effectively combined with encoders (BanglaBERT, XLM-RoBERTa) for Bangla sentiment analysis?

RQ2: What is the relative performance trade-off between topic modeling + encoder pipelines and direct LLM prompting?

RQ3: How can we design a cost-aware evaluation framework that jointly considers accuracy, latency, energy consumption, and carbon emissions?

RQ4: Under uncertainty (e.g., fluctuating costs, variable energy prices), which models remain robust in sentiment analysis tasks?

This study defines a set of research objectives that directly align with the formulated research questions. The first objective (RO1) is to develop a comprehensive comparative framework evaluating 36 distinct model configurations for Bangla sentiment analysis. This objective responds to RQ1 and RQ2 by systematically comparing how traditional topic modeling approaches (e.g., LDA, BERT-LDA, BERTopic) and LLM-based topic modeling can be effectively combined with pre-trained encoders (such as BanglaBERT and XLM-RoBERTa), as well as by contrasting these pipelines against direct LLM prompting. The second objective (RO2) focuses on integrating robust optimization techniques into model evaluation in order to account for operational and sustainability costs. This objective is directly linked to RQ3, which asks how we can design a cost-aware evaluation framework that jointly considers accuracy, latency, energy consumption, and carbon emissions. The third objective (RO3) is to analyze trade-offs between interpretability,

modularity, scalability, and efficiency across different model families. This directly builds upon RQ2 and RQ4 by examining how different models balance performance with transparency and adaptability, especially under uncertain conditions such as fluctuating computational costs or variable energy availability. Finally, the fourth objective (RO4) aims to establish a benchmark for Bangla-specific sentiment analysis that extends beyond conventional accuracy metrics. In line with RQ3 and RQ4, this benchmark will incorporate broader evaluation dimensions, including time efficiency, energy consumption, and environmental sustainability, thereby providing a holistic foundation for future studies in Bangla sentiment analysis.

3. Methodology

This section outlines the methodological framework that will be adopted in the proposed study. The overall process will consist of data collection and preprocessing, model development across three comparative paradigms (classical embedding-based topic modeling, LLM-based approaches, and direct prompting), evaluation through accuracy, efficiency, and sustainability metrics, and integration of robust optimization techniques to address uncertainty in model performance and operational costs. Figure 1 provides a schematic overview of the proposed methodology. The methodology is intended to benchmark 36 model configurations based on their accuracy, efficiency, and sustainability.

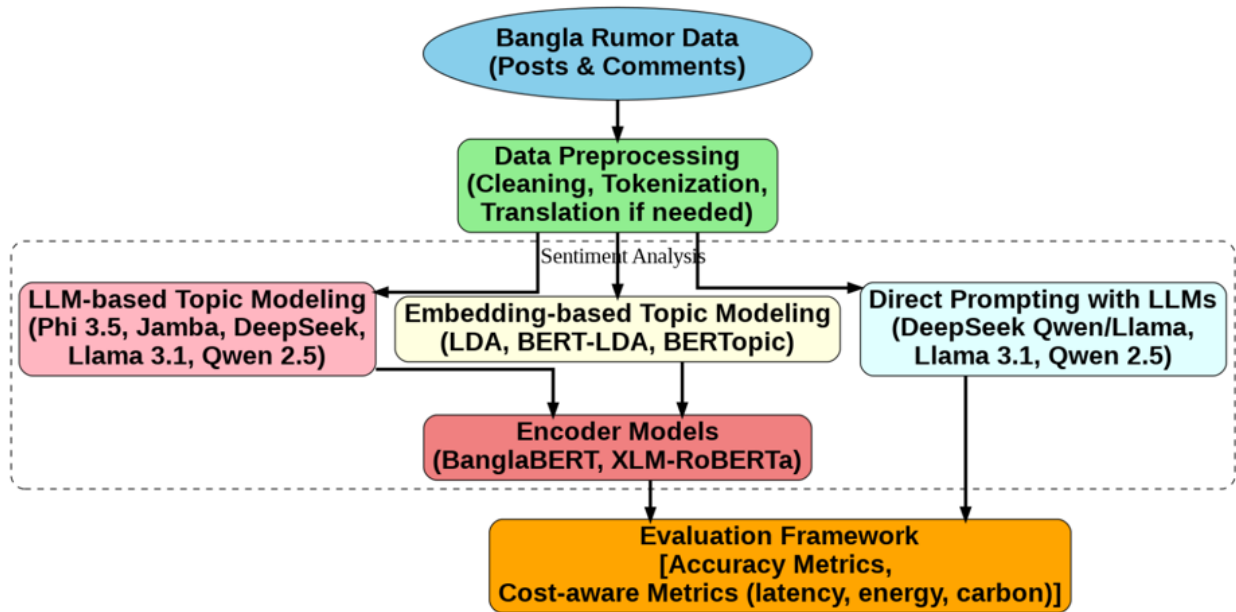


Figure 1: Flowchart of the proposed methodology.

3.1 Data Extraction and Preprocessing

The dataset that will be employed in this study will consist of Bangla rumor-related posts and comments collected from Facebook. A semi-automatic data collection pipeline will be implemented. First, rumor-related links will be extracted from the Rumor Scanner platform, which serves as an organized repository of misinformation cases in the Bangladeshi context. These links will form the basis for automated collection, whereby a custom web scraper (e.g., Apify) will be developed to retrieve both the original post content and the associated user comments.

Following collection, the dataset will undergo a systematic preprocessing procedure to ensure its suitability for natural language processing tasks. Noise elements, including duplicates, hyperlinks, emojis, and irrelevant symbols, will be removed to preserve the linguistic integrity of the corpus. The cleaned text will then be tokenized, which means that sentences will be broken down into Bangla-specific tokens that machine learning models can use later on. After that, a normalization process will be used to fix spelling mistakes and other typical informal spellings. Finally, code-mixed segments containing both Bangla and English expressions will be standardized through translation into a unified Bangla representation. This multi-stage preprocessing pipeline will ensure the creation of a standardized and linguistically coherent corpus, forming a robust foundation for subsequent sentiment and topic modeling experiments. An illustrative example of the extracted dataset, including post titles, user comments, and corresponding stance labels, is presented in Table 3.

Table 3: sample data

Post Title	Comments	Stance
ইন্দোনেশিয়া কোরআনের হাফেজ কে মাটি দেয়ার সময় কবর আলোকিত হয়ে যায়। সুবহানাল্লাহ্ (In Indonesia, the grave of a Hafiz of the Quran lit up during the burial. SubhanAllah.)	সুবহানাল্লাহ্ (SubhanAllah.)	Positive
	মিথ্যা কথা (False statement.)	Negative

	দয়া করে আমার পোস্ট টা আপনাদের গ্রুপে ছারেন (Please share my post in your group.)	Neutral
--	---	---------

3.2 Topic Modeling

Topic modeling constitutes a key methodology for uncovering latent semantic patterns in text corpora and plays a vital role in sentiment classification. In this study, two broad categories of topic modeling are considered: embedding-based topic modeling and LLM-based topic modeling (Table 4).

Table 4: Embedding-Based and LLM-Based Topic Modeling Approaches

Embedding-based topic modeling	LDA
	BERT-LDA
	BERTopic
LLMs for prompting-based topic modeling	Phi 3.5 Mini / MoE
	Jamba Mini 1.5
	DeepSeek distillations (DS Qwen 14B / 32B, DS Llama 8B / 70B)
	Llama 3.1 8B / 70B
	Qwen 2.5 7B / 72B

Embedding-based approaches, such as Latent Dirichlet Allocation (LDA), BERT-LDA, and BERTopic, represent classical and hybrid methods for topic discovery. LDA operates as a generative probabilistic model, assuming that each document is a mixture of latent topics, while each topic is represented as a distribution over words. This unsupervised and language-agnostic design makes it particularly suitable for low-resource settings like Bangla, where large annotated datasets are often unavailable. However, LDA is limited by its inability to capture semantic meaning, context, or word order, and its performance tends to degrade on short, noisy texts. Extensions such as BERT-LDA and BERTopic attempt to overcome these shortcomings by incorporating contextual embeddings from pre-trained models. This integration improves both the semantic coherence of discovered topics and their interpretability for downstream sentiment analysis tasks.

LLM-based approaches leverage recent advancements in large-scale foundation models, such as Phi 3.5, Jamba Mini, DeepSeek distillations (Qwen, Llama), Llama 3.1, and Qwen 2.5. These models enable topic extraction through prompting rather than relying on fixed probabilistic assumptions. By capturing rich semantic dependencies, contextual nuances, and multilingual patterns, LLM-based topic modeling offers significant advantages for Bangla sentiment analysis, particularly in handling informal expressions, code-switching, and culturally embedded cues. Nevertheless, these benefits come at a cost: LLMs typically require substantial computational resources, result in higher inference latency, and consume more energy, raising concerns about their sustainability and scalability.

The comparison of fundamentals, assumptions, advantages, and limitations across these approaches is summarized in Table 5. While embedding-based models excel in efficiency, transparency, and low-resource adaptability, LLM-based models provide richer semantic representations and flexibility but face constraints in terms of cost and sustainability. Together, these categories provide complementary perspectives: embedding-based methods emphasize interpretability and computational feasibility, whereas LLM-based methods highlight contextual depth and adaptability.

Table 5: Comparison of Topic Modeling Approaches: Fundamentals, Advantages, and Limitations

Model	Fundamentals & Assumptions	Advantages	Limitations
LDA (Latent Dirichlet Allocation)	- Generative probabilistic model. - Each document = mixture of latent topics; each topic = distribution over words. - Assumes bag-of-words independence (word order ignored).	- Unsupervised and language-agnostic – requires no annotated data, making it suitable for low-resource languages like Bangla. - Interpretable topic-word distributions. - Stronger generalization than pLSI/LSA via Dirichlet priors.	- Ignores semantics, context, and word order. - Requires predefined number of topics. - Performs poorly on short texts.
BERT-LDA (BERT + LDA Hybrid)	Uses contextual embeddings (BERT) to improve semantic coherence.	Addresses LDA’s limitation: lack of semantics.	Resource-intensive (embedding + clustering).

	' Assumes embeddings capture meaning better than raw co-occurrence.	' Produces more coherent topics (groups synonyms/related phrases). ' Better performance on short texts.	' Still needs bag-of-words for topic labeling.
BERTopic (Embedding-based Topic Model)	' Clusters transformer embeddings of documents. ' Uses class-based TF-IDF (c-TF-IDF) for topic labels.	' Addresses BERT-LDA's complexity: integrates clustering + labeling in one pipeline. ' High topic coherence. ' Modular (any embedding model can be plugged in). ' Handles dynamic topics over time.	' Assigns one dominant topic per document. ' Still bag-of-words for topic labels → misses phrase-level nuance. ' Heavy computation for huge datasets.
Phi 3.5 (Mini & MoE)	' Small language models (SLMs) with transformer backbone. ' <i>Mini</i>: efficient, 3.8B params, 128k context. ' <i>MoE</i>: 16 experts, ~42B total, ~6.6B active per inference.	' Addresses BERTopic's limitation: non-generative, limited reasoning. ' Strong reasoning/coding ability despite small size. ' MoE balances performance vs efficiency.	' Bangla is not explicitly listed; for Bangla, it might occasionally produce awkward phrasing or rely on more general knowledge. ' Smaller models still lack deep world knowledge. ' MoE adds deployment complexity.
Jamba 1.5 Mini (AI21Labs)	' Hybrid architecture: Transformer + State-Space Model (SSM). ' Supports 256k context (longest among peers).	' Addresses Phi-3.5's limitation: limited context. ' State-of-the-art long document analysis. ' Faster decoding efficiency for long inputs.	' Bangla was not explicitly listed as a supported language. ' Complex architecture (MoE + SSM). ' Very high hardware requirements for large models.
DeepSeek Distillations (DS Qwen-14B/32B &	' Knowledge distilled from a huge 671B MoE teacher into smaller Qwen/Llama bases.	' Efficient reasoning in multiple languages (like Bangla). ' Addresses Jamba's limitation: weaker	' Bias and flaws from the teacher model can transfer.

DS Llama-8B/70B)	' Focus on reasoning and chain-of-thought.	' reasoning compared to distilled teacher knowledge. ' Superior reasoning at smaller scale. ' Performs step-by-step problem solving well.	' Still large (8B–70B), not lightweight.
Llama 3.1 (8B / 70B)	' Meta's dense LLM is trained on 15T tokens. ' Instruction-tuned with 10M curated examples. ' 128k context.	' High general intelligence and strong multilingual abilities (for its supported languages) ' Addresses DeepSeek's limitation: narrow focus on reasoning. ' Balanced general-purpose model. ' Strong ecosystem + fine-tuning support.	' Resource-heavy (esp. 70B). ' Slightly behind Qwen in multilingual/code benchmarks.
Qwen 2.5 (7B / 72B)	' Trained on 18T tokens (largest among peers). ' Strong multilingual + coding specialization. ' 32k context.	' State-of-the-art multilingual performance and multimodality: Qwen-2.5-72B is one of the most capable open models as of 2025. ' Addresses Llama 3.1's limitation: not top in multilingual/code tasks. ' State-of-the-art open-source performance. ' Excels at reasoning, coding, and multilingual tasks.	' 72B model requires heavy computing. ' Proprietary MoE versions only via API.

3.3 Sentiment Classification

Sentiment classification is a core task in natural language processing that determines whether text expresses positive, negative, or neutral opinions. In Bangla, this task is particularly challenging

due to limited annotated datasets, frequent use of *Banglish* (Bangla–English code-mixing), and transliterated writing styles. To address these challenges, this study considers both **encoder-based models** and **LLM-based models** for sentiment classification (Table 6).

Table 6: Encoder-Based and LLM-Based Models for Sentiment Classification

Encoder-Based	BanglaBERT
	XLM-RoBERTa
LLM-Based	DeepSeek distillations (DS Qwen 14B / 32B, DS Llama 8B / 70B)
	Llama 3.1 8B / 70B
	Qwen 2.5 7B / 72B

Encoder-Based Models, such as BanglaBERT is a monolingual BERT-style encoder trained exclusively on Bangla corpora using masked language modeling. By focusing on Bangla-only data, it captures finer syntactic and semantic structures than multilingual encoders, enabling high accuracy in Bangla sentiment classification. Its compact size (~110M parameters) makes it efficient for fine-tuning and deployment. However, BanglaBERT struggles in real-world contexts where Banglish and transliterated Bangla are common, and its fixed 512-token limit restricts long text processing. Moreover, as an encoder-only model, it lacks generative or reasoning capabilities. On the other hand, XLM-RoBERTa (XLM-R), a multilingual RoBERTa model, offers broader coverage by being trained on more than 100 languages. Unlike BanglaBERT, it can handle Banglish and transliterated input more effectively, making it more versatile for noisy social media data. Its multilingual nature also supports cross-lingual transfer learning. However, this breadth comes at the cost of depth—XLM-R generally underperforms BanglaBERT in tasks requiring deep Bangla-specific linguistic understanding.

LLM-Based Models, such as DeepSeek distillations (Qwen 14B/32B, Llama 8B/70B), Llama 3.1 (8B/70B), and Qwen 2.5 (7B/72B) extend sentiment classification by enabling direct prompting. These models excel at capturing semantic richness, contextual meaning, and subtle sentiment cues like sarcasm or idiomatic expressions, which are common in Bangla discourse. They also offer few-shot and zero-shot capabilities, reducing dependence on annotated Bangla datasets. Despite these strengths, their high computational cost, inference latency, and energy consumption make them less sustainable for large-scale or resource-constrained deployments. Furthermore, their

effectiveness depends heavily on the amount and quality of Bangla data included in pre-training, which is often limited compared to high-resource languages.

The comparison of fundamentals, assumptions, advantages, and limitations across these approaches is summarized in Table 7. Encoder-based models offer lightweight, specialized, and fine-tunable solutions for Bangla sentiment tasks, while LLM-based models provide flexible, context-rich, and prompt-driven alternatives at the expense of efficiency and sustainability. A balanced evaluation of these models requires not only conventional **accuracy-based comparisons** but also **cost-aware considerations**, including latency, energy consumption, and environmental impact. This study adopts such a unified evaluation framework to benchmark encoder and LLM-based approaches for Bangla sentiment classification, providing a holistic perspective that reflects both performance and real-world feasibility.

Table 7: Comparison of Sentiment Classification Approaches: Fundamentals, Advantages, and Limitations

Model	Fundamentals & Assumptions	Advantage	Limitations
BanglaBERT	- Monolingual BERT-base encoder trained only on Bangla text. - Uses masked language modeling. - Assumes Bangla-only training captures better syntax and semantics than multilingual.	- Directly overcomes the limitation of generic multilingual models that lacked Bangla depth. - Strong grasp of Bangla vocabulary and grammar. - Higher accuracy on Bangla sentiment tasks. - Lightweight (~110M params) and easy to fine-tune.	- Poor at handling Banglish (Bangla + English mix). - Struggles with transliterated Bangla. - Limited to 512 tokens. - No generative or reasoning ability.
XLM-RoBERTa	Multilingual RoBERTa encoder trained on 100 languages (including Bangla).	Overcomes BanglaBERT’s limitation of poor handling of code-mixing and transliteration.	- Bangla is underrepresented in its training → misses idioms/slang. - Still context-limited (~512 tokens).

	' Uses shared subword vocabulary for cross-lingual transfer. ' Assumes high-resource languages can boost low-resource languages.	' Covers Bangla plus related languages, improving mixed text handling.	' Not tuned specifically for Bangla sentiment out of the box.
DeepSeek Distillations (DS Qwen-14B/32B & DS Llama-8B/70B)	' Distilled models using Qwen or Llama backbones. ' Knowledge distilled from reasoning-heavy training. ' Assumes reasoning skill transfers to multiple languages.	' Overcomes XLM-R's limitation of limited capacity and short context. ' Longer context windows → handles full Bangla threads or articles. ' Strong reasoning → can detect sarcasm or irony in Bangla sentiment. ' Efficient open-source models easier to adapt than massive dense LMs.	' Not natively trained on Bangla → needs extra fine-tuning. ' Larger models still require significant compute. ' Misses some idiomatic Bangla cues unless adapted.
Llama 3.1 (8B / 70B)	' Next-gen Transformer (8B–405B params). ' Trained on large multilingual corpora, with 128K context. ' Assumes massive scale + instruction tuning = superior multilingual performance.	' Overcomes DeepSeek's limitation of limited Bangla exposure. ' Trained on broader language data → stronger Bangla fluency. ' Detects subtle sentiment shifts and sarcasm in Bangla. ' Can analyze entire long documents or conversations.	' Very large (70B+ params) → costly to run. ' Bangla still less represented than English/Chinese. ' Improvements in sentiment classification may be marginal over smaller tuned models.
Qwen 2.5 (7B / 72B)	' Alibaba's efficient LLM family (up to 72B params). ' Long context (128K), optimized training to rival larger models.	' Overcomes Llama 3.1's limitation of impractical scale and deployment. ' Nearly state-of-the-art Bangla understanding at smaller size.	' Training still focused mainly on Chinese/English → Bangla needs fine-tuning. ' Subword tokenization may

	Assumes efficiency + smart training can outperform larger dense models.	Long context helps with lengthy sentiment-rich Bangla texts. Easier to fine-tune and deploy than 70B–405B Llama models.	fragment Bangla words. 72B model still resource-demanding for small labs.
--	---	---	---

3.4 Comparative Framework for Sentiment Analysis

The proposed framework will employ three distinct paradigms for Bangla sentiment analysis: (i) Classical topic modeling + encoders, (ii) LLM-based Topic Modeling + Encoders, and (iii) Direct LLM prompting. These approaches will enable a systematic comparison of interpretability, modularity, efficiency, and scalability.

3.4.1 Classical Topic Modeling + Encoders

Embedding-based topic modeling will be implemented by integrating classical probabilistic techniques with modern encoder representations. Latent Dirichlet Allocation (LDA) will be employed as a baseline generative model for latent topic discovery, while BERT-LDA will combine contextual embeddings with probabilistic topic assignment. In addition, BERTopic will be applied to cluster transformer-based embeddings into coherent topic structures. These approaches will leverage pre-trained Bangla encoders such as BanglaBERT and XLM-RoBERTa to generate semantically rich representations, thereby enhancing topic coherence.

3.4.2 LLM-based Topic Modeling + Encoders

State-of-the-art LLMs, including Phi 3.5, Jamba, DeepSeek, Llama 3.1, and Qwen 2.5, will be prompted to generate interpretable topics directly, without relying on explicit clustering algorithms. Their embeddings will then be integrated with encoder-based classifiers to predict sentiment polarity.

Although state-of-the-art LLMs (e.g., DeepSeek, Llama 3.1, Qwen 2.5) can be directly prompted for sentiment classification, this study also adopts a LLM-based topic modeling with encoder integration approach. A key reason behind this choice is the structure of the dataset, where the **number of posts is much smaller than the number of comments**. Directly applying LLMs for classification on such a large-scale comment set would require substantial computational resources, resulting in higher carbon emissions and longer processing times. By contrast, using LLMs for

topic modeling and then feeding the resulting embeddings into lightweight encoder-based classifiers provides a more **efficient and sustainable solution**. This approach not only reduces environmental impact but also enables a fair comparison with embedding-based topic modeling pipelines, ensuring methodological consistency across models.

3.4.3 Direct LLM Prompting

In contrast to encoder-based approaches, direct prompting will bypass the topic modeling stage entirely. Zero-shot and few-shot prompting strategies will be employed with the aforementioned LLMs to classify Bangla texts into sentiment categories. This paradigm will serve as a baseline for evaluating trade-offs among accuracy, modularity, and computational cost.

Table 8: Combinations for Sentiment Analysis

No.	Combination for Sentiment Analysis	
	Topic Modeling	Sentiment Classification
1	LDA	BanglaBERT
2	LDA	XLM-RoBERTa
3	BERT-LDA	BanglaBERT
4	BERT-LDA	XLM-RoBERTa
5	BERTopic	BanglaBERT
6	BERTopic	XLM-RoBERTa
7	Phi 3.5 Mini	BanglaBERT
8	Phi 3.5 Mini	XLM-RoBERTa
9	Phi 3.5 MoE	BanglaBERT
10	Phi 3.5 MoE	XLM-RoBERTa
11	Jamba Mini 1.5	BanglaBERT
12	Jamba Mini 1.5	XLM-RoBERTa
13	DeepSeek DS Qwen 14B	BanglaBERT
14	DeepSeek DS Qwen 14B	XLM-RoBERTa
15	DeepSeek DS Qwen 32B	BanglaBERT
16	DeepSeek DS Qwen 32B	XLM-RoBERTa
17	DeepSeek DS Llama 8B	BanglaBERT
18	DeepSeek DS Llama 8B	XLM-RoBERTa
19	DeepSeek DS Llama 70B	BanglaBERT
20	DeepSeek DS Llama 70B	XLM-RoBERTa
21	Llama 3.1 8B	BanglaBERT
22	Llama 3.1 8B	XLM-RoBERTa
23	Llama 3.1 70B	BanglaBERT

24	Llama 3.1 70B	XLM-RoBERTa
25	Qwen 2.5 7B	BanglaBERT
26	Qwen 2.5 7B	XLM-RoBERTa
27	Qwen 2.5 72B	BanglaBERT
28	Qwen 2.5 72B	XLM-RoBERTa
29	-	DeepSeek DS Qwen 14B
30	-	DeepSeek DS Qwen 32B
31	-	DeepSeek DS Llama 8B
32	-	DeepSeek DS Llama 70B
33	-	Llama 3.1 8B
34	-	Llama 3.1 70B
35	-	Qwen 2.5 7B
36	-	Qwen 2.5 72B

By implementing these three comparative tracks, the study will generate a benchmark encompassing 36 distinct model configurations in Table 8. Table 8 outlines 36 experimental configurations across three categories: (i) traditional topic modeling with encoders (No. 1–6), (ii) LLM-based topic modeling with encoders (No. 7–28), and (iii) direct LLM prompting (No. 29–36). The first group uses classical models (LDA, BERT-LDA, BERTopic) with BanglaBERT or XLM-RoBERTa. The second uses LLMs (e.g., Phi 3.5, Jamba, DeepSeek, Llama 3.1, Qwen 2.5) to generate topic embeddings before encoder-based sentiment classification. The third skips topic modeling, applying zero/few-shot prompting of LLMs directly. This structure enables systematic benchmarking of interpretability, modularity, and computational efficiency.

3.5 Evaluation Metrics

Traditional sentiment analysis studies primarily rely on **usual performance metrics** such as accuracy, precision, recall, F1-score, and confusion matrix. While these metrics are useful for measuring classification quality, they present a key limitation: they fail to account for **time, energy consumption, carbon footprint, and the cost of misclassification**. As a result, model evaluation remains incomplete when applied to large-scale and real-world scenarios where efficiency and sustainability are critical.

To address this limitation, we adopt a **robust optimization framework** that extends beyond conventional accuracy-based measures. Specifically, our evaluation incorporates four additional cost dimensions:

- **Latency** ($\alpha_T T_j$) – the time required for inference, weighted by latency importance (Strubell et al., 2020).
- **Energy** ($\alpha_E E_j$) – the energy consumed during computation, weighted by energy cost factors (Strubell et al., 2020).
- **Carbon footprint** ($\alpha_C C_j$) – the greenhouse gas emissions measured in CO₂-equivalents, scaled by environmental cost weights (Lannelongue et al., 2021).
- **Cost-sensitive errors** ($\alpha_{\text{err}} \sum_{i \neq j} c_{ij} N_{ij}$) – the penalty for misclassification, where costs differ across error types (Elkan, 2001).

By integrating these factors, the proposed framework ensures that model selection is **robust under uncertainty** (e.g., fluctuating costs, variable energy prices, carbon pricing). This approach supports a **holistic evaluation** that jointly captures **accuracy, efficiency, sustainability, and impact**, offering a more realistic basis for model comparison in Bangla sentiment analysis.

3.5 Robust Optimization Formulation

To account for uncertainty in evaluation priorities, this study adopts a robust optimization framework that integrates accuracy, efficiency, sustainability, and cost-sensitive errors into a unified decision-making formulation. To formally define the robust optimization framework for sentiment analysis, the notations are represented in Table 9.

Table 9: Explanation of notations used in Robust Optimization.

- $\alpha_T, \alpha_E, \alpha_C, \alpha_{\text{err}}$: uncertain weights (time, energy, carbon, error cost per unit)
- $[\underline{\alpha}, \bar{\alpha}]$: lower and upper bounds of uncertainty interval
- m : model candidate (e.g., BERT, DistilBERT, etc.)
- $T(m)$: latency (seconds)
- $E(m)$: energy consumption (kWh)
- $C_{\text{COe}}(m)$: carbon footprint (kg COe)
- x_j : binary variable (1 if model j is selected, else 0)
- $N_{ij}(m)$: confusion matrix entries (misclassifications of class i as j)
- c_{ij} : cost of misclassifying i as j
- M : total number of models in candidate set
- $\mathcal{U}$: uncertainty set (all coefficient ranges)

The goal is to identify model configurations that remain optimal under worst-case variations in evaluation weights. The formulation proceeds in four steps.

Step 1: Interval Definitions

The weights assigned to evaluation criteria—accuracy (α_T), efficiency/energy (α_E), sustainability/carbon (α_C), and misclassification penalties (α_{err})—are not assumed to be fixed. Instead, each weight is modeled as belonging to a bounded interval:

$$\alpha_T \in [\underline{\alpha}_T, \bar{\alpha}_T], \quad \alpha_E \in [\underline{\alpha}_E, \bar{\alpha}_E], \quad \alpha_C \in [\underline{\alpha}_C, \bar{\alpha}_C], \quad \alpha_{\text{err}} \in [\underline{\alpha}_{\text{err}}, \bar{\alpha}_{\text{err}}]$$

This setup captures uncertainty in stakeholder preferences (e.g., shifting emphasis from accuracy to sustainability).

Step 2: Robust (Min–Max) Formulation

Given a set of models M , the performance of model $m \in M$. The robust objective seeks the configuration that minimizes the worst-case weighted score:

$$\min_{m \in \mathcal{M}} \max_{\alpha \in \mathcal{U}} \left[\alpha_T T(m) + \alpha_E E(m) + \alpha_C C_{COe}(m) + \alpha_{err} \sum_{i \neq j} c_{ij} N_{ij}(m) \right]$$

where $T(m)$ is the accuracy score, $E(m)$ is energy consumption, $C_{coe}(m)$ is the carbon footprint, and $N_{ij}(m)$ denotes misclassifications from class i to j , penalized by c_{ij} .

Step 3: MILP Formulation (Discrete Model Selection)

To operationalize this selection, binary decision variables x_j are introduced, where

$$x_j = \begin{cases} 1 & \text{if model } j \text{ is chosen} \\ 0 & \text{otherwise} \end{cases}$$

Where $x_j=1$ if model j is chosen and $x_j = 0$ otherwise. The optimization model becomes:

$$\min_x \max_{\alpha \in \mathcal{U}} \sum_{j=1}^M x_j \left(\alpha_T T_j + \alpha_E E_j + \alpha_C C_j + \alpha_{err} \sum_{i \neq k} c_{ik} N_{ik,j} \right)$$

subject to:

$$\sum_{j=1}^M x_j = 1, \quad x_j \in \{0, 1\}$$

This ensures exactly one model is selected from the set of candidates.

Step 4: Linearization (Robust Counterpart)

The robust counterpart replaces each uncertain weight with its worst-case bound, yielding a tractable linear objective:

$$\max_{\alpha \in \mathcal{U}} \left(\alpha_T T_j + \alpha_E E_j + \alpha_C C_j + \alpha_{err} \sum_{i \neq k} c_{ik} N_{ik,j} \right) = \bar{\alpha}_T T_j + \bar{\alpha}_E E_j + \bar{\alpha}_C C_j + \bar{\alpha}_{err} \sum_{i \neq k} c_{ik} N_{ik,j}$$

This linearized version is computationally efficient and ensures that the chosen configuration remains **optimal under worst-case realizations** of the evaluation weights, thereby guaranteeing robustness in model selection.

3.6 Trade-off Analysis

Following robust optimization, a structured trade-off analysis will be conducted to evaluate how different model configurations balance competing objectives. Models will be compared along axes such as **interpretability versus scalability** and **accuracy versus sustainability**, enabling an assessment of the practical implications of model choice. Visualization techniques, including scatter plots, radar charts, and Pareto frontiers, will be employed to identify configurations that achieve a desirable compromise among multiple criteria. This analysis provides actionable insights for selecting models that satisfy both technical and operational priorities in the Bangla sentiment analysis context.

4. Data collection plan

The dataset for this study is composed of Bangla rumor-related posts and comments from Facebook. A semi-automatic pipeline, including link extraction from the Rumor Scanner platform and web scraping with Apify, is employed for collection. The full procedure, along with preprocessing steps, has been described in detail in Section 3.1 (Data Extraction and Preprocessing).

5. Conclusion

5.1 Summary

This study establishes a benchmark of 36 experimental configurations for Bangla sentiment prediction, spanning traditional topic modeling with encoders, LLM-based topic modeling with encoders, and direct LLM prompting. The results are expected to show that LLM-based topic modeling produces richer embeddings, improving classification accuracy compared to classical methods. BanglaBERT is anticipated to outperform XLM-RoBERTa in Bangla-only tasks due to its language-specific pretraining, while direct LLM prompting may yield competitive performance but at higher latency and energy costs. The proposed cost-aware evaluation framework highlights

that the most suitable model depends not only on accuracy but also on deployment constraints such as efficiency, scalability, and sustainability. Overall, the study contributes both theoretical insights and practical guidelines for sustainable sentiment analysis in low-resource languages.

5.2 Limitations

Despite the comprehensive design of the proposed framework, several limitations remain. First, the current study focuses solely on sentiment prediction and does not extend to related tasks such as stance detection (support, deny, query, comment), which could provide deeper insights into misinformation dynamics. Second, while large-scale LLM prompting demonstrates strong performance, it incurs high computational and energy costs, limiting its feasibility for real-time applications. Future work could address this by exploring knowledge distillation or parameter-efficient tuning techniques (e.g., LoRA, adapters) to reduce inference costs for Bangla text. Third, the framework primarily relies on text-based features, without incorporating graph-based or multimodal signals such as images, videos, or network propagation patterns, which are often critical in social media rumor analysis. Finally, the evaluation is conducted on a curated dataset rather than fully evolving real-world misinformation events. Future research should extend testing to live Bangla social media datasets across platforms such as Facebook, Twitter, and Reddit. Additionally, there is potential to develop an open-source benchmark toolkit for cost-aware sentiment prediction in low-resource languages, supporting broader adoption and reproducibility.

6. Reference

- Ahmed, M. F., Mahmud, Z., Biash, Z. T., Ryen, A. A. N., Hossain, A., & Ashraf, F. B. (2021). Cyberbullying detection using deep neural network from social media comments in bangla language. *arXiv preprint arXiv:2106.04506*.
- Ahmed, M. T., Rahman, M., Nur, S., Islam, A., & Das, D. (2021, February). Deployment of machine learning and deep learning algorithms in detecting cyberbullying in bangla and romanized bangla text: A comparative study. In *2021 International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT)* (pp. 1-10). IEEE.
- Akbik, A., Bergmann, T., Blythe, D., Rasul, K., Schweter, S., & Vollgraf, R. (2019, June). FLAIR: An easy-to-use framework for state-of-the-art NLP. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics (demonstrations)* (pp. 54-59).
- Birjali, M., Kasri, M., & Beni-Hssane, A. (2021). A comprehensive survey on sentiment analysis: Approaches, challenges and trends. *Knowledge-Based Systems*, 226, 107134.
- Das, A. K., Al Asif, A., Paul, A., & Hossain, M. N. (2021). Bangla hate speech detection on social media using attention-based recurrent neural network. *Journal of Intelligent Systems*, 30(1), 578-591.
- Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019, June). Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)* (pp. 4171-4186).
- Emon, E. A., Rahman, S., Banarjee, J., Das, A. K., & Mitra, T. (2019, June). A deep learning approach to detect abusive bengali text. In *2019 7th International Conference on Smart Computing & Communications (ICSCC)* (pp. 1-5). IEEE.

- Eshan, S. C., & Hasan, M. S. (2017, December). An application of machine learning to detect abusive bengali text. In *2017 20th International conference of computer and information technology (ICCIT)* (pp. 1-6). IEEE.
- Gandhi, U. D., Malarvizhi Kumar, P., Chandra Babu, G., & Karthick, G. (2021). Sentiment analysis on twitter data by using convolutional neural network (CNN) and long short term memory (LSTM). *Wireless Personal Communications*, 1-10.
- Haque, R., Islam, N., Tasneem, M., & Das, A. K. (2023). Multi-class sentiment classification on Bengali social media comments using machine learning. *International journal of cognitive computing in engineering*, 4, 21-35.
- Hutto, C., & Gilbert, E. (2014, May). Vader: A parsimonious rule-based model for sentiment analysis of social media text. In *Proceedings of the international AAAI conference on web and social media* (Vol. 8, No. 1, pp. 216-225).
- Ishmam, A. M., & Sharmin, S. (2019, December). Hateful speech detection in public facebook pages for the bengali language. In *2019 18th IEEE international conference on machine learning and applications (ICMLA)* (pp. 555-560). IEEE.
- Jain, P. K., Pamula, R., & Srivastava, G. (2021). A systematic literature review on machine learning applications for consumer sentiment analysis using online reviews. *Computer science review*, 41, 100413.
- Ligthart, A., Catal, C., & Tekinerdogan, B. (2021). Systematic reviews in sentiment analysis: a tertiary study. *Artificial intelligence review*, 54(7), 4997-5053.
- Lin, L., & Chen, Z. (2021). Social rumor detection based on multilayer transformer encoding blocks. *Concurrency and Computation: Practice and Experience*, 33(6), e6083.
- Loria, S. (2018). textblob Documentation. *Release 0.15*, 2(8), 269.
- Mathew, L., & Bindu, V. R. (2020, March). A review of natural language processing techniques for sentiment analysis using pre-trained models. In *2020 Fourth international conference on computing methodologies and communication (ICCMC)* (pp. 340-345). IEEE.

- Mitra, A., & Mohanty, S. (2021). Sentiment analysis using machine learning approaches. In *Emerging Technologies in Data Mining and Information Security: Proceedings of IEMIS 2020, Volume 2* (pp. 63-68). Singapore: Springer Nature Singapore.
- Patterson, D., Gonzalez, J., Le, Q., Liang, C., Munguia, L. M., Rothchild, D., ... & Dean, J. (2021). Carbon emissions and large neural network training. *arXiv preprint arXiv:2104.10350*.
- Paul, O., & Yesmin, S. (2024). Rethinking the Misinformation with its Detrimental Impact on Lives: A Qualitative Approach. *Science & Technology Libraries*, 43(3), 251-266.
- Pennington, J., Socher, R., & Manning, C. D. (2014, October). Glove: Global vectors for word representation. In *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)* (pp. 1532-1543).
- Pirayani, R., Madhavi, D., & Singh, V. K. (2017). Analytical mapping of opinion mining and sentiment analysis research during 2000–2015. *Information Processing & Management*, 53(1), 122-150.
- Politico. After Charlie Kirk's killing, false claims flourish online — with help from US adversaries. Available online: <https://www.politico.com/news/2025/09/17/charlie-kirk-shooting-disinformation-russia-china-iran-00570357> (accessed on 17 September 2025)
- Sagi, S. (2025). Optimizing LLM Inference: Metrics that Matter for Real Time Applications. *Journal of Artificial Intelligence & Cloud Computing*. SRC/JAICC-475. DOI: doi.org/10.47363/JAICC/2025 (4), 446, 2-4.
- Wang, P., Shi, H., Wu, X., & Jiao, L. (2021, September). Sentiment analysis of rumor spread amid COVID-19: Based on weibo text. In *Healthcare* (Vol. 9, No. 10, p. 1275). MDPI.
- Yadav, A., & Vishwakarma, D. K. (2020). Sentiment analysis using deep learning architectures: a review. *Artificial Intelligence Review*, 53(6), 4335-4385.
- Yousif, A., Niu, Z., Tarus, J. K., & Ahmad, A. (2019). A survey on sentiment analysis of scientific citations. *Artificial Intelligence Review*, 52(3), 1805-1838.
- Gaikwad, A., Belhekar, P., & Kottawar, V. (2023, December). Advancing Multilingual Sentiment Understanding with XGBoost, SVM, and XLM-RoBERTa. In *International Conference on*

- Data Science, Machine Learning and Applications (pp. 990-1000). Singapore: Springer Nature Singapore.
- Ghatora, P. S., Hosseini, S. E., Pervez, S., Iqbal, M. J., & Shaukat, N. (2024). Sentiment analysis of product reviews using machine learning and pre-trained llm. *Big Data and Cognitive Computing*, 8(12), 199.
- Grootendorst, M. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv preprint arXiv:2203.05794*.
- Iftikhar, S., & Davy, S. (2024, November). Reducing carbon footprint in ai: A framework for sustainable training of large language models. In *Proceedings of the Future Technologies Conference* (pp. 325-336). Cham: Springer Nature Switzerland.
- Invernici, F., Curati, F., Jakimov, J., Samavi, A., & Bernasconi, A. (2025). Capturing research literature attitude towards sustainable development goals: an LLM-based topic modeling approach. *Journal of Big Data*, 12(1), 139.
- Katta, K. (2025). Analyzing User Perceptions of Large Language Models (LLMs) on Reddit: Sentiment and Topic Modeling of ChatGPT and DeepSeek Discussions. *arXiv preprint arXiv:2502.18513*.
- Khan, T., Motie, S., Kocak, S. A., & Raza, S. (2025). Optimizing Large Language Models: Metrics, Energy Efficiency, and Case Study Insights. *arXiv preprint arXiv:2504.06307*.
- Kowsher, M., Sami, A. A., Prottasha, N. J., Arefin, M. S., Dhar, P. K., & Koshiba, T. (2022). Bangla-bert: transformer-based efficient model for transfer learning and language understanding. *IEEE Access*, 10, 91855-91870.
- Liu, V., & Yin, Y. (2024). Green AI: exploring carbon footprints, mitigation strategies, and trade offs in large language model training. *Discover Artificial Intelligence*, 4(1), 49.
- Miah, M. S. U., Kabir, M. M., Sarwar, T. B., Safran, M., Alfarhood, S., & Mridha, M. F. (2024). A multimodal approach to cross-lingual sentiment analysis with ensemble of transformer and LLM. *Scientific Reports*, 14(1), 9603.

- Miah, M. S. U., Kabir, M. M., Sarwar, T. B., Safran, M., Alfarhood, S., & Mridha, M. F. (2024). A multimodal approach to cross-lingual sentiment analysis with ensemble of transformer and LLM. *Scientific Reports*, 14(1), 9603.
- Pandya, S. (2024). Comparative Analysis of Large Language Models and Traditional Methods for Sentiment Analysis of Tweets Dataset. *Int. J. Innov. Sci. Res. Technol*, 9(12), 1647-1657.
- Rehan, M., Malik, M. S. I., & Jamjoom, M. M. (2023). Fine-tuning transformer models using transfer learning for multilingual threatening text identification. *IEEE Access*, 11, 106503-106515.
- Różycki, R., Solarska, D. A., & Waligóra, G. (2025). Energy-Aware Machine Learning Models—A Review of Recent Techniques and Perspectives. *Energies*, 18(11), 2810.
- Sagi, S. (2025). Optimizing LLM Inference: Metrics that Matter for Real Time Applications. *Journal of Artificial Intelligence & Cloud Computing*. SRC/JAICC-475. DOI: doi.org/10.47363/JAICC/2025 (4), 446, 2-4.
- Thakur, N., Cui, S., Knieling, V., Khanna, K., & Shao, M. (2024). Investigation of the misinformation about covid-19 on youtube using topic modeling, sentiment analysis, and language analysis. *Computation*, 12(2), 28.
- Uddin, M. A., & Sarker, I. H. (2024). An explainable transformer-based model for phishing email detection: A large language model approach. *arXiv preprint arXiv:2402.13871*.
- van der Veen, A. M., & Bleich, E. (2025). The advantages of lexicon-based sentiment analysis in an age of machine learning. *PloS one*, 20(1), e0313092.
- Wahidur, R. S., Tashdeed, I., Kaur, M., & Lee, H. N. (2024). Enhancing zero-shot crypto sentiment with fine-tuned language model and prompt engineering. *IEEE Access*, 12, 10146-10159.
- Wahidur, R. S., Tashdeed, I., Kaur, M., & Lee, H. N. (2024). Enhancing zero-shot crypto sentiment with fine-tuned language model and prompt engineering. *IEEE Access*, 12, 10146-10159.
- Wang, Z., Zhang, T., Zhu, L., Bai, H., Hou, L., Xiang, S., ... & Liu, W. (2025). Faster and better llms via latency-aware test-time scaling. *arXiv preprint arXiv:2505.19634*.

- Xing, F. (2025). Designing heterogeneous llm agents for financial sentiment analysis. *ACM Transactions on Management Information Systems*, 16(1), 1-24.
- Xu, S., Wu, Z., Zhao, H., Shu, P., Liu, Z., Liao, W., ... & Li, X. (2024). Reasoning before comparison: LLM-enhanced semantic similarity metrics for domain specialized text analysis. *arXiv preprint arXiv:2402.11398*.
- Yates, A., Nogueira, R., & Lin, J. (2021, March). Pretrained transformers for text ranking: BERT and beyond. In *Proceedings of the 14th ACM International Conference on web search and data mining* (pp. 1154-1156).
- Strubell, E., Ganesh, A., & McCallum, A. (2020, April). Energy and policy considerations for modern deep learning research. In *Proceedings of the AAAI conference on artificial intelligence* (Vol. 34, No. 09, pp. 13693-13696).
- Lannelongue, L., Grealey, J., & Inouye, M. (2021). Green algorithms: quantifying the carbon footprint of computation. *Advanced science*, 8(12), 2100707.
- Elkan, C. (2001, August). The foundations of cost-sensitive learning. In *International joint conference on artificial intelligence* (Vol. 17, No. 1, pp. 973-978). Lawrence Erlbaum Associates Ltd.
- Islam, K. I., Islam, M. S., & Amin, M. R. (2020, December). Sentiment analysis in Bengali via transfer learning using multi-lingual BERT. In *2020 23rd International Conference on Computer and Information Technology (ICCIT)* (pp. 1-5). IEEE.
- Hoque, M. N., & Seddiqui, M. H. (2024, March). Exploring transformer ensemble approach to classify cyberbullying text for the low-resource bengali language. In *2024 International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS)* (pp. 1-6). IEEE.
- Hoque, M. N., & Seddiqui, M. H. (2023, December). Leveraging transformer models in the cyberbullying text classification system for the low-resource bengali language. In *2023 26th International Conference on Computer and Information Technology (ICCIT)* (pp. 1-6). IEEE.
- Amin, I., Haque, E., Noor, M. N., Ahmmed, M. M., Puri, S., & Babu, M. A. (2024, April). From Detection to Disruption: Leveraging NLP Insights to Proactively Combat Cybercrime in

Bangladesh. In International Conference on Innovations in Computational Intelligence and Computer Vision (pp. 81-95). Singapore: Springer Nature Singapore.